09: Intro to Multiple Linear Regression

Stat 230 - Spring 2026

> **Note**
>
> This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use https://aluby.github.io/stat230/activities/09-mlr

1 Data

- In an effort to understand the important aspects of a satisfactory supervisor, clerical employees at a large financial organization were asked to rate their immediate supervisor.
- The survey questions were designed to measure overall performance of the supervisor, as well as six additional characteristics.
- Employees were asked to rate the following statements on a scale from 0 to 100 (0 = "completely disagree", 100 = "completely agree")

Note: this data comes from the textbook

Variable	Description
`overall`	Overall rating of supervisor performance
`complaints`	Score for "Your supervisor handles employee complaints appropriately."
`privileges`	Score for "Your supervisor allows special privileges."
`learn`	Score for "Your supervisor provides opportunities to learn new things."
`raises`	Score for "Your supervisor bases raises on performance."
`critical`	Score for "Your supervisor is too critical of poor performance."
`advance`	Score for "I am not satisfied with the rate I am advancing in the company?"

Before looking at the data, how do you think each predictor variable will be related to `overall`? What do you think will have the strongest relationship? The weakest?

1.1 Loading data

To load the `supervisor` data set, run the following code chunk:

```
library(dplyr)
library(GGally)

supervisor <- read.csv("https://aluby.github.io/stat230/data/supervisor.csv")
```

1.2 EDA: Making a pairs plot

We will use the `ggpairs()` function in the {GGally} package to make the pairs plot. By default, it will include every variable in the dataset. The code below firsts `select()`s the columns we want to include (all of them). I have also moved the response variable to be the last in the list, so that it shows up on the Y axis.

```
supervisor |>
  select(complaints, privileges, learn, raises, critical, advance, overall) |>
  ggpairs()
```

- Which predictor variables appear to be most closely related to `overall`?
- Which predictor variables appear to be most closely related to *each other*?

1.3 SLR

In the code chunk below, choose what you think will be the best *single* predictor and fit an SLR model. Interpret the intercept and slope coefficient and report the R^2 value.

```
supervisor_slr <- lm(overall ~ ______, data = supervisor)
summary(supervisor_slr)
```

1.4 MLR

In the code chunk below, use all predictors and fit an MLR model. Interpret the intercept and two of the slope coefficients and report the R^2 value.

```
supervisor_mlr <- lm(overall ~ ______, data = supervisor)
summary(supervisor_mlr)
```

2 Predicting responses

We can use most our SLR tools for MLR. For example, here's some code I could use to calculate predictions. (Uses the `bikes` data)

```
predict(bikes_lm, newdata = data.frame(temp_actual = 90, windspeed = 6))
```

- Use this as starter code to predict the overall manager score for a manager with scores of 50% for all predictors
- What if the manager from the previous question decides to try to improve her `critical` score by 10 points (so 40% instead of 50%) and her `complaints` score by 10 points (so 60% instead of 50%). How does that change the prediction?

```
# your code here
```

2.1 Function quick reference

Function	Purpose
`lm(formula, data)`	Fit a MLR model
`predict(model, newdata, interval, level, se.fit)`	Make predictions and calculate intervals
`data |> select(vars) |> ggpairs()`	Make pairs plot for selected variables